1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

Dr Samit Bhattacharya
Computer Science and Engineering
IIT Guwahati

2 **L17 – Case Study on Prototype**

Dr Samit Bhattacharya
Computer Science and Engineering
IIT Guwahati

3 **Recap**

- We learned about prototypes
  - What are those
  - How to make
  - Different types

4 **Recap**

- We also learned about prototype evaluation
  - Cognitive walkthrough
  - Heuristic evaluation

5 **Recap**

- In this lecture
  - Learn about prototype building and evaluation through a case study

6 **The Case**

7 **Calendar App Case**

- A calendar app – introduced earlier
  - Recap - we are interested to build a calendar app. It is meant for the students (mainly) to help in their various academic activities

8 **SRS**

- Already discussed earlier
- Remember, SRS is the starting point for code design (our concern here)

9 **SRS – Recap (Wk 3, L8)**

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Prototype with Cognitive Walkthrough

11 **Overview**

- A medium fidelity, vertical prototype
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- Prototype built with figma tool

12 **Overview**

- prototype has
  - Months in a Year View

- Days in a Month View
- Specific Date Views
- Set Reminder View
- Edit Reminder View
- Delete Reminder View
- Set Theme View
  - SelectThemeView
  - ThemeAppliedViews

13 **Months in a Year View**

14 **Days in a Month View**

15 **Specific Date Views**

16 **Specific Date Views**

17 **Specific Date Views**

18 **Set Reminder View**

19 **Set Reminder View**

20 **Edit Reminder View**

21 **Edit Reminder View**

22 **Delete Reminder View**

23 **Select Theme View**

24 **Theme Applied Views**

25 **Theme Applied Views**

26

Use Cases for Cognitive Walkthrough

27 **Use Case 1**

- You have been given an assignment in the Software Engineering Lab course. To avoid missing the deadline, you would like to add a reminder for the deadline of the assignment

28 **Use Case 1**

- Interface level tasks
  - Navigate to the date of the deadline, by choosing the correct year and month, in case the month is different from the current month
  - Select the date of the deadline
  - Select the + button
  - Add the description of the assignment, its type as deadline and the time of deadline
  - Save

29 **Use case 1**

30 **Questions for use Case 1**

- Were you able to navigate to the correct month easily?
  - Expert 1: Yes, I was able to navigate by swiping left from the current month
  - Expert 2: Yes, since the month being displayed was the month in which I had to set the reminder
  - Expert 3: Yes, a simple swipe on the screen showed the next month's calendar. This is a common way to move to the next element in mobile applications
  - Expert 4: Yes, I was able to navigate to the correct month easily

31 **Questions for use Case 1**

- Did you navigate to the wrong month by mistake? If yes, was the mechanism to go to the correct month convenient?
  - Expert 1: Yes, but it was easy to go to the correct month by swiping
  - Expert 2: No, I selected the correct month
  - Expert 3: No, I did not navigate to the wrong month. I was able swipe correctly on the first attempt
  - Expert 4: No I did not navigate to the wrong month

32 **Questions for use Case 1**

- Was it apparent that you could add a reminder by clicking on the date?
  - Expert 1: Though it wasn't that apparent, I did click the date to add a reminder on that date. The events listed below the dates helped. However, I am not sure about how to add events in an effective manner which would span across more than a day.
  - Expert 2: Yes, not at first but I figured it out.
  - Expert 3: It was not completely apparent, but because there wasn't any other option
  - for it in the default view, it was extremely likely.
  - Expert 4: It was not apparent that I could add a reminder by clicking on the date, I just happened to click on a date and was directed to the view that contained an option to add the reminder.
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33 **Questions for use Case 1**

- Did the + button indicate its function appropriately?
  - Expert 1: Yes, the + button made it clear that it is meant for adding a new event to the calendar.
  - Expert 2: Yes, it was apparent that it displayed some functionality to add something, which could be nothing other than reminders.
  - Expert 3: Yes, it was apparent that the + button was to add reminders.
  - Expert 4: Yes. The + button indicated its functionality properly. I was able to easily add a reminder.
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34 **Questions for use Case 1**

- Were the input fields in the set reminder window clearly visible and was their naming appropriate?
  - Expert 1: The input fields were clearly visible. However, for adding the deadline, both start and end was shown which confused me. I would have liked to have just an end time of the deadline.
  - Expert 2: Yes, the input fields were appropriately labeled and were visible.
  - Expert 3: Yes, the input fields were colored differently to the background. The labels
  - made it easy to understand the purpose of each input field.
  - Expert 4: Yes. The input fields in the set reminder were visible and their names clearly reflected what they were meant for.
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35 **Questions for use Case 1**

- Were you able to see the assignment after addition in the deadline view?
  - Expert 1: Yes, I was able to see the assignment added in the deadline view and it was also highlighted in the calendar. The ”saved successfully” message also confirmed the addition of the assignment.
  - Expert 2: Yes, the deadline appeared in the deadline view with the banner displaying that it was added successfully.
  - Expert 3: I could see the event added correctly in the specific day view.
  - Expert 4: Yes. The assignment was visible in the deadline view after addition, I also liked that the assignments were sorted by time.
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36 **Use Case 2**

- Due to some unavoidable circumstances, your class for the Software Engineering course today has been shifted to 10:00 am to 11:00 am. You would like to edit the reminder for the class with updated timings

37 **Use Case 2**

- Interface level tasks
  - Select today’s date
  - Select the Software Engineering class event
  - Edit the start and end time fields to 10:00 am and 11:00 am respectively
  - Save

38 **Use case 2**

39 **Questions for use Case 2**

- Were you able to locate today’s date?
  - Expert 1: Yes, it was highlighted in blue color
  - Expert 2: Yes, the corresponding current day cell had highlighted background in the month’s view
  - Expert 3: Yes, the current day’s date had been highlighted in a different color to make it stand out
  - Expert 4: Yes, today’s date was clearly highlighted I had no difficulty in locating it

40 **Questions for use Case 2**

- Was it apparent that you could edit the time of the class by clicking on the event?
  - Expert 1: Yes, it was clear that I'll have to open that event first to make any changes to it
  - Expert 2: Yes, not at first but playing around made it clear that clicking on an event had some functionality associated with it, which was to edit and delete
  - Expert 3: While it was not certain, it was highly probable because most mobile applications allow editing elements by selecting them
  - Expert 4: Editing the time of the class by clicking on the event was slightly ambiguous, the tiles representing the events did not look like buttons and an edit option was not explicitly mentioned

41 **Questions for use Case 2**

- Was it apparent that the fields were editable?
  - Expert 1: It wasn't apparent from the fields itself. But the window popped up showing edit reminder made it clear that I can edit the fields
  - Expert 2: Yes, clicking the fields made them turn into edit mode, then it could be edited
  - Expert 3: The "Edit Reminder" heading at the top made me confident that the fields here were editable. This was confirmed when I clicked on the input field
  - Expert 4: Yes it was apparent that the fields were editable , as the fields either closely resembled text boxes or radio buttons

42 **Questions for use Case 2**

- Did you make an error while editing and saving? If yes, was the mechanism to correct it convenient?
  - Expert 1: No, I was able to correctly edit the time fields and save it.
  - Expert 2: No, I correctly edited the fields and saved successfully.
  - Expert 3: Yes, while editing the event; I deleted the event by mistake. So, I had to make a new event with the same details. It would be better if you could give a warning, or ask for confirmation before deletion.
  - Expert 4: No. I did not make an error while editing and saving.

43 **Questions for use Case 2**

- Were you able to see the class event with updated timings?
  - Expert 1: Yes, I was able to see the "Updated successfully" message, along with the update in the timings of the event
  - Expert 2: Yes, it was visible in the events view for a particular date, and banner was displayed showing a successful update.
  - Expert 3: Yes, after saving the edited reminder, the class timings had been updated in other views too.
  - Expert 4: Yes I was able to see the class event with updated timings, the green pop up indicating that the update is successful made the process transparent.

44 **Use Case 3**

- You want to meet your friends during the third week next month. You want to check your schedule to find an appropriate date and time for the meet

45 **Use Case 3**

- Interface level tasks
  - Swipe left in the current month to navigate to the next month
  - Get an overview of your schedule for the third week
  - Select the date in which you have the least number of events scheduled.
  - Check the timings of the events, and decide a convenient time to meet your friend.

46 **Use Case 3**

47 **Questions for Use Case 3**

- Were you able to navigate to the next month without going to the year's view?
  - Expert 1: Yes, I was able to figure out that you could go to the next month by swiping left.
  - Expert 2: No, the swipe left feature was not so apparent at first, so initially going to month's view seemed to be the only option.
  - Expert 3: Yes, I did not need to go the year view. However, initially, I was not very sure how to go to the next month; and I needed some time to figure it out.
  - Expert 4: Yes I was able to navigate to the next month without going to the months view, however the I feel that the transition could be add more apparent by using a next or a forward button in place of the right swipe mechanism being currently used.

48 **Questions for Use Case 3**

- Were the events listed clearly visible in the month's view?
  - Expert 1: No, I felt that the font size could have been increased. I noticed that not all events where shown in the month.
  - Expert 2: No, not all the events were listed, only first three in chronological order were visible from day-wise view.
  - Expert 3: The first few (3) events of each day could be seen clearly. However their timings had not been given.
  - Expert 4: Yes the events were clearly visible. The events were also color coded which gave a rough overview of the kind and number of the events scheduled on a particular day.
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49 **Questions for Use Case 3**

- Were you able to determine the day with the least number of events effectively through the month's view?
  - Expert 1: Yes, the month's view gave a good idea of which day would be relatively free without the need to see the events in each day of the week.
  - Expert 2: Yes, but in my case I chose the day with only one event.
  - Expert 3: Yes. An overview of each day's schedule was visible, with which I could
  - make out the day with least events scheduled.

- Expert 4: Yes. Task headings for the tasks that needed to be performed on a particular day were clearly specified this allowed me to efficiently determine the days with the least number of events.
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50 **Questions for Use Case 3**

- Were you able to decide a convenient time to meet your friend which doesn't clash with the other events in the day?
  - Expert 1: Yes, after viewing the events in the day with least number of events, I was able to decide a convenient time. The events being displayed in the sorted order of time helped.
  - Expert 2: Yes, I chose 12:00 hrs to 16:00 hrs slot on 18th April as it had the least number of events.
  - Expert 3: Yes, I could see the timings of all events on that day, and decide an appropriate time accordingly.
  - Expert 4: Yes , I could easily decide a convenient time to meet with my friend on a given day. The events scheduled on a given day were clearly specified and were sorted by time, this allowed me to efficiently decide a convenient time to meet with my friend
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51 **Use Case 4**

- You want to change the theme of the calendar with some different background image
- Interface level tasks
  - Click on the figure icon, which is labeled as "Theme"
  - Select the theme that you wish to apply

52 **Use Case 4**

53 **Questions for Use Case 4**

- Were you able to set theme correctly?
  - Expert 1:Yes, the icons were labeled appropriately, rest of the procedure was simple
  - Expert 2:Yes, I could easily set the theme
  - Expert 3:Yes, I applied the theme without any problems
  - Expert 4:Yes, I could easily apply the theme

54 **General Questions about the Prototype**

- Was there any component in the interface which you could not understand?
  - Expert 1: No, I was able to understand all the components of the interface as they have clear functions
  - Expert 2: Most of the things were fairly easy to grasp other than the year view where the months in a year were displayed. The swiping action though was not apparent at first but is fairly easy to grasp.
  - Expert 3: The swipe method to change to adjacent months and years was not understandable at first. Also, changing to year view from month view by clicking on the year was not clearly visible. Editing a reminder was also not very clear at first. However, when I used the app for some time, I was able to understand each

component
- Expert 4: I was able to understand all the components in the interface. For the large part, the interface was intuitive and easy to use and the functions that were performed by individual components were clear
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55 **General Questions about the prototype**
- Was sufficient feedback provided indicating that intended action(s) was/were performed?
  - Expert 1:Yes, the feedback was provided after each action instantly. I was also able to see the updates in the application correctly after the action.
  - Expert 2:Yes, feedback was provided showing updated, deleted and saved reminder banners. In case of setting themes, the theme was applied to the background.
  - Expert 3:Yes, feedback was given wherever required. Whenever a reminder was saved, edited or deleted; a banner confirming the same was shown. For other actions too, the changes caused by the action were visible to the user.
  - Expert 4:Sufficient feedback was provided after the intended actions were performed. For instance whenever I added/deleted/updated a reminder feedback related to the action was shown in a green prompt, this allowed me to easily keep track of what was going on.

56 **General Questions about the Prototype**
- Was the interface in accordance with the conventions of the real world applications?
  - Expert 1:Yes, the interface was in accordance with the real world conventions. The icons used and their color followed the conventions. The positioning of the buttons also was in accordance. It was similar to most of the standard calendar applications
  - Expert 2:Yes, the days were listed appropriately, with events. The event view was as found in different applications. Its adherence to common conventions makes it easier to use
  - Expert 3:Yes, it was. "Delete" and "Deadline" buttons were shown with the "red danger" color, while "Save" button and confirmations of successful operations were shown in green. The "+" sign was used to indicate adding reminders
  - Expert 4:The interface was in accordance conventions of the real world applications .Since I had a prior experience using such applications I found the given application very easy to use. General conventions such as displaying deadlines using red color, using the plus ('+') symbol in the button for deadline addition were followed which made the application intuitive
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57 **General Questions about the Prototype**
- Was the design appropriate for a calendar application?
  - Expert 1:Yes, the design was appropriate and the lighter background seemed good. However, the font size on the month view could have been increased. Overall, the design was neat, clean and simple.
  - Expert 2:Yes, the design follows a normal calendar, be it the month view, year

view. It is as one would find in a real calendar.
- Expert 3:Yes, the design was appropriate, with different views for a day, month and year making it convenient for the user to view the calendar at whatever level he wished.
- Expert 4:Overall the design is appropriate for a calendar application. Certain aspects of the design such as displaying the deadlines sorted by time , giving feedback after actions made the application easy to use. However at some points the design was not very intuitive and improvements could be made. For instance , in the view where deadlines are shown, it is not very clear that I can click on the deadline and update it because the deadline does not resemble a button. Similarly it was not apparent that I can switch from month view to year view by clicking on the year because the year did not resemble a button
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Result Analysis

59 **Result and Analysis**

- Each evaluator was able to perform the tasks smoothly and achieve the required objective. They were all satisfied with the overall experience.
- The controls for essential actions such as addition/deletion/updation of events were provided through buttons which reflected their intended purpose.
- Whenever the user performed some action, the app went through visible changes. The feedback was provided by the app to the user, whenever the user added, edited or deleted a reminder
- Although errors were made by the evaluators at some point, they were able to recover from the error. One evaluator faced an error when he mistakenly deleted the reminder, and had to go through the process of adding the reminder again. It would be better if we could show a confirmation prompt before saving or deleting a reminder
- The interface is similar to most of the other calendar applications. At some points, the steps to achieve the desired functionality were not apparent, but the evaluators' experience with other similar applications guided them.
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60 **Analysis for Functionality**

- Navigating Between Various Months and Years
  - The procedure to go to the year view by clicking on the current year is not apparent to some users
  - The swiping method to go the next month in the month view, and the next year in the year view, was difficult to find for some users
  - Once in the year view, one can then go the desired month by simply clicking on it, which is natural

61 **Analysis for Functionality**

- Viewing the events of a specific day
  - The procedure to view the events of a specific day is quite intuitive and clear

- By simply clicking on the day in the month view, we can view all events of that day
- These events are sorted in order of their start time. Further, clearly labeled buttons give the user the option of seeing a particular type of event (timetable / deadline / meeting)
- Each type of event is assigned its own color. So the user can easily make out the type by looking at the event

62 **Analysis for Functionality**

- Add/Edit/Delete Events
  - The '+' button for adding events gives the user a clear indication of its purpose
  - The procedure for editing/deleting events, though, is not very obvious
  - Users did not feel confident that clicking on the event would enable them to perform those operations
  - Once in the add/edit reminder view, users can easily save the events with appropriate details
  - It would be better to ask for confirmation before saving changes, to handle errors

63 **Analysis for Functionality**

- Set Theme
  - The button to set the theme is clearly visible in the main view, and is well labeled. In the next step, the user can easily select the desired theme
  - The process is fairly simple for a layman user